



BAAZ & UDAY JKBOSE

2026 CLASS 11TH

SETS

MATHS

LECTURE-04

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TOPICS TO BE COVERED

1

Subsets

2

Intervals as Subsets of $\mathbb{R}$

(not Discussed Today)

2

Related Problems

Subsets

- A set B is said to be a Subset of a Set A if every element of B is contained in A.

Generally denoted by $B \subseteq A$

(Subset of A) (Superset of B)

For example: $B = \{2, 4, 6\}$ and $A = \{1, 2, 4, 6, 8\}$

$\Rightarrow B \subseteq A$

$B = \{2, 1, 4\}$ and $A = \{1, 3, 4, 5\}$

$\Rightarrow B \not\subseteq A$

Q) What is the number of Subsets of a Set A?

* ϕ is the Subset of every set.

* Every Set is a Subset of Itself.

* $A = \{1, 2\}$

List of subsets of A:

$\phi, A, \{1\}, \{2\}$

Total no. of Subsets of $A = 4 = 2^2$.

* $A = \{1, 2, 3\}$.

List of Subsets of A :

$\phi, A, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$

$\Rightarrow$ Total no. of Subsets of $A = 8 = 2^3$.

** If Cardinality of $A = n$, then the number of Subsets of $A = 2^n$

Question: If $A = \{1, 2, 3, 4\}$, then the number of Subsets of A is:

a) 8 (b) 4 ~~(c) 16~~ (d) 32.

no. of Subsets $= 2^4 = 16$.

Note:

- * ϕ & A are known as Trivial Subsets of A .
- * A is known as the Improper Subset of A .
- * Any other Subset B ($B \neq A$) of A is known as Proper Subset of A .

* $B \subseteq A \Rightarrow x \in A \forall x \in B$

$\forall \rightarrow$ for all.

* $B \not\subseteq A \Rightarrow$ there exists $x \in B$ but $x \notin A$.

* **Power Set**: • Set of Subsets of A .

• Generally denoted by $P(A)$.

• $P(A) = \{x : x \subseteq A\}$

e.g.: • $A = \{1\}$
 $\Rightarrow P(A) = \{\phi, A\}$

• $A = \{1, 2\}$.

$\Rightarrow P(A) = \{\phi, A, \{1\}, \{2\}\}$

Question: If $A = \{1, 2, 3, 4\}$, then the Cardinality of $P(A)$ is:

~~(a) 8~~

~~(b) 16~~

~~(c) 24~~

~~(d) 32~~

$n(P(A)) = 2^4 = 16.$

QUESTION

The set $\{1, 2\}$ is the proper subset of

~~1~~

$\{1, 2\}$ ✗

~~2~~

$\{1, 3, 4\}$ ✗

~~3~~

$\{1, 2, 3\}$ ✓

~~4~~

$\{\{1, 2\}, 3, 4\}$ ✗

QUESTION

The set $A = \{1, 2, 3\}$ is the subset of

~~1~~ $\{1, 2, 4, 5\}$ ✗

~~2~~ $\{1, \{2, 3\}, 4, 5\}$ ✗

~~3~~ $\{1, 2, 3, 7, 8\}$ ✓

~~4~~ $\{\{1, 2\}, 3, 4\}$

QUESTION

If set $A = \{1, 2, 3\}$, then $P(A)$ is equal to

- ☒ 1 $\{\phi, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}\}$
- ☒ 2 $\{\{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{3,1\}\}$
- ☒ 3 $\{\phi, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\}\}$
- ☒ 4 $\{\{1\}, \{2\}, \{3\}, \{1,2,3\}\}$

QUESTION

Which of the Following is **True**?

~~1~~

$a \in \{\{a\}, b\}$ ✗

~~2~~

$\{b, c\} \subset \{a, \{b, c\}\}$ ✗

~~3~~

$\{a, b\} \subset \{a, \{b, c\}\}$

~~4~~

None of these ✓

$$\{b, c\} \in \{a, \{b, c\}\}$$
$$\{\{b, c\}\} \subset \{a, \{b, c\}\}$$

✓ $A = \{\{1\}, 2\}$

$\{1\}, 2$

* $A = \{1, \{2, 3\}\}$

$1, \{2, 3\}$

• $x \in A \Rightarrow \{x\} \subset A$

• $\{x\} \in A \Rightarrow \{\{x\}\} \subset A$

QUESTION (NCERT EXEMPLAR):

$$a^x = a^y \Rightarrow x = y \quad a \neq 0, -1, 1.$$



Two finite sets have m and n elements. The number of subsets of the first set is 112 more than that of the second set. The values of m and n are, respectively.

~~1~~ 4, 7

☒ 2 7, 4

~~3~~ 4, 4

~~4~~ 7, 7

A $\rightarrow$ first set $n(A) = m$

B $\rightarrow$ Second set $n(B) = n$

$$2^m = 2^n + 112$$

$$\Rightarrow 2^3 - 2^2 = 112$$

$$\Rightarrow 2^2(2^{m-n} - 1) = 112$$

$$\Rightarrow 2^{\text{even}} (2^{\text{odd}} - 1) = 2^4 \times 7$$

So on Comparing, we get;

$$2^n = 2^4 \Rightarrow n = 4.$$

$$2^{m-n} - 1 = 7$$

$$\Rightarrow 2^{3-4} = 8$$

$$\Rightarrow 2^{m-4} = 2^3$$

$$\Rightarrow m - 4 = 3.$$

$$\Rightarrow m = 7$$

2	112
2	56
2	28
2	14
	7

QUESTION (JEE MAINS 2020):

Set A has m elements and set B has n elements. If the total number of subsets of A is 112 more than the total subsets of B, then the value of $m.n$ is _____.

~~(a) 14~~ ~~(b) 28~~ ~~(c) 35~~ ~~(d) 42~~

$$\begin{aligned} m &= 7 \\ n &= 4 \Rightarrow mn = 7 \times 4 = 28. \end{aligned}$$

Universal Set

Set which contains all the sets under consideration.

e.g: $A = \{1, 2\}$, $B = \{3, 4\}$, $C = \{5, 6\}$

$$U = \{1, 2, 3, 4, 5, 6\}$$

$$A \subset U, B \subset U, C \subset U.$$

QUESTION

Which of the following sets is the Universal set of sets $A = \{2, 4, 5\}$, $B = \{1, 3, 5\}$, $C = \{3, 5, 7, 11\}$, $D = \{2, 4, 8, 10\}$

~~1~~ $\{1, 2, 3, 5, 6, 7, 8, 9, 10\}$ ~~X~~

~~2~~ $\{1, 2, 3, 4, 5, 6, 7, 8, 10\}$ ~~X~~

☒ 3 $\{1, 2, 3, 4, 5, 6, 7, 8, 10, 11\}$ ✓

4 $\{1, 2, 3, 4, 6, 7, 8, 10, 11\}$ ~~X~~

* NCERT

* NCERT EXAMPLER.

* objective R.D. Sharma

Jan-Mehraj1729.

$$2^n = 2^4$$

$$n = 4.$$

$$a^x = a^y$$

$$x = y, a \neq 0, -1.$$

Thank
You